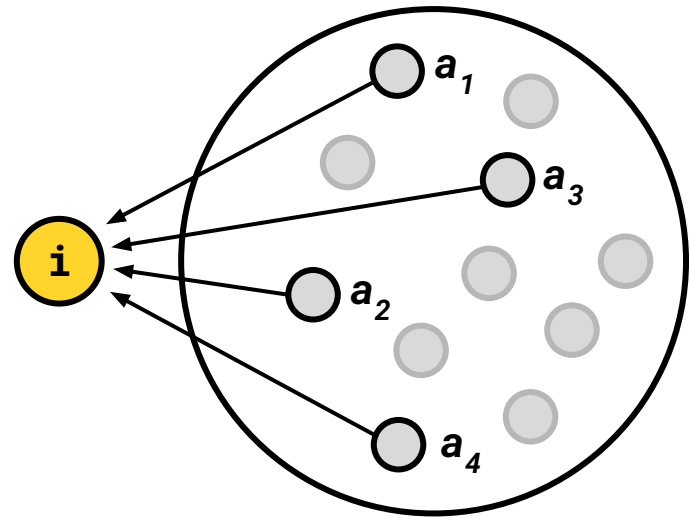


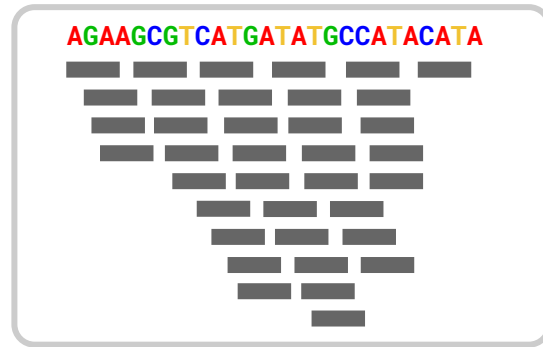
A. Exome read depth modeling

Exome samples



Reference set (R_i) = $\{a_1, a_2, a_3, a_4\}$

exon 'j'



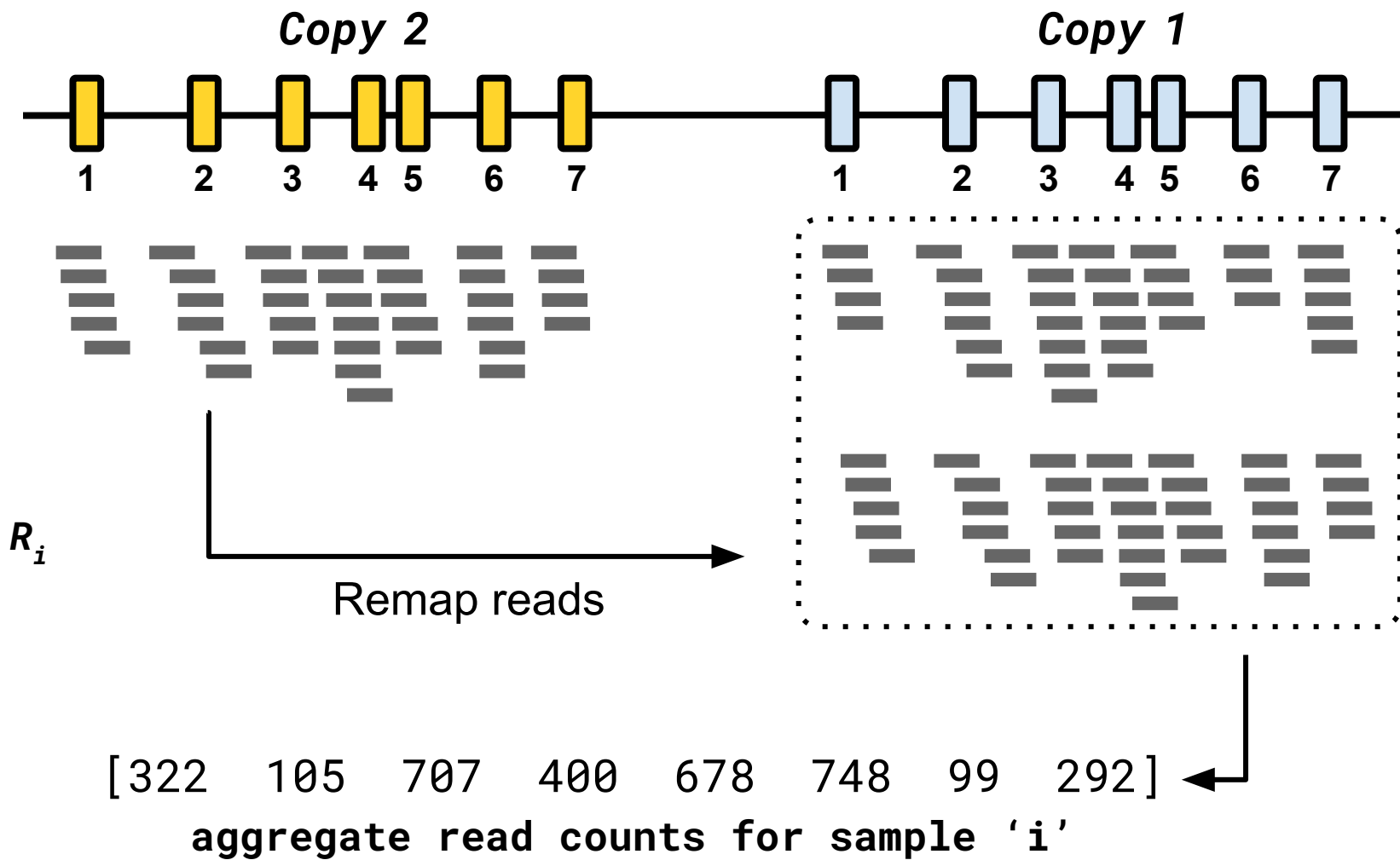
d_{ij} = read count for sample i

$D_{ij} = d_{ij} + \text{read count for samples in } R_i$

$d_{ij} \sim \text{Beta-binomial}(D_{ij}, \alpha, \beta)$

Expected read count ratio for each exon (no CNV) = $\alpha:\beta$

B. Aggregate read counting



C. Aggregate copy number estimation

